Graphify Report — Cargo Delivery Service

Generated: 2026-06-14
Repository: D: **Analysis Scope:** Full codebase (architecture docs, frontend source, mock data, status mapping engine)

Executive Summary

The Cargo Delivery Service is a centralized multi-carrier logistics platform. It unifies 4 major carriers (DHL, FedEx, UPS, Chronopost) under a single adapter interface with a sophisticated status mapping engine that translates 64+ carrier-specific status codes into 12 unified internal codes across 3 matching strategies (exact, regex, fuzzy). The frontend is a React + Vite + Tailwind TypeScript SPA with full CRUD on all entities, JWT-mock authentication, and multilingual support (EN/FR/TR).

The graph contains **40 nodes** and **48 edges** organized into **4 communities**.

Graph Statistics

Metric	Value
Total Nodes	40
Total Edges	48
Communities	4
EXTRACTED edges	43 (89.6%)
INFERRED edges	5 (10.4%)
Avg. confidence score	0.949

Community Map

Community 0 — Architecture & Carrier Adapters

Color: Blue (#3b82f6)
Nodes: 7

The architectural backbone. A unified `CarrierAdapter` interface is implemented by 4 carrier-specific adapters (DHL, FedEx, UPS, Chronopost), each normalizing carrier-specific concepts into a shared `UnifiedStatusModel`. The `CARGO-SERVICE-ARCHITECTURE.md` document defines the full contract.

- **God Node:** `cargo_service_arch` (degree: 5) — references endpoint docs, defines carrier adapter, connects to all adapter nodes
- **Key Pattern:** Pluggable adapter pattern — adding a new carrier requires only implementing the interface

Community 1 — Status Mapping Engine

Color: Green (#10b981)
Nodes: 6

The most technically sophisticated component. `StatusMapperEngine` maintains per-carrier maps with 3-tier matching (exact string → regex patterns → fuzzy substring fallback). It detects terminal vs. blocking states, builds milestone timelines, and verifies lifecycle transitions.

Carrier	Codes Mapped	Strategy Count
DHL	15	Exact (8) + Regex (5) + Fuzzy (2)
FedEx	16	Exact (9) + Regex (5) + Fuzzy (2)
UPS	18	Exact (10) + Regex (6) + Fuzzy (2)
Chronopost	15	Exact (7) + Regex (5) + Fuzzy (3)

- **God Node:** `status_mapper` (degree: 6) — central hub connecting to all carrier maps + milestone builder + unified status model
- **God Node:** `unified_status` (degree: 5) — used by all 4 adapters and the mapper

Community 2 — Frontend Application

Color: Yellow (#f59e0b)
Nodes: 21

The largest community. A React 18 + TypeScript SPA built with Vite and Tailwind CSS, organized around a service-layer architecture.

Layer Architecture:

```
Pages (dashboard, shipments, carriers, etc.)
  → Shared Components (Sidebar, DataTable, Modal, etc.)
    → React Hooks (useApi, useMutation)
      → Cargo Service Layer (20+ API functions)
        → Mock Data (6 entity stores)
        → Status Mapper Engine
```

Key Metrics: - 9 page components, all with full CRUD modals - 6+ shared/reusable components - 4 form components (ShipmentForm, CarrierForm, PickupForm, ServiceForm) - AuthGuard route protection wrapping all pages - 750+ i18n keys across 3 locales

God Nodes: - `react_frontend` (degree: 6) — root dependency hub - `cargo_service_layer` (degree: 9) — most highly connected node in the graph, used by every page - `auth_system` (degree: 3) — crosscuts both frontend and REST endpoint design

Community 3 — API Design & Documentation

Color: Purple (#8b5cf6)
Nodes: 7

Documents the RESTful API contract covering 18 endpoints organized around 6 functional areas: - Shipment Lifecycle (7 states: draft → cancelled/delivered) - Carrier CRUD (add/edit/delete carriers) - Rate Engine (quote calculation) - Pickup Flow (schedule/cancel pickups) - Webhook Simulation (trigger/receive webhooks) - Address Validation

God Node: `rest_endpoints` (degree: 6) — documents all functional areas

God Nodes (High-Connectivity Hubs)

Node	Community	Degree	File Type
cargo_service_layer	Frontend	9	code
cargo_service_arch	Architecture	5	document
unified_status	Architecture	5	code
status_mapper	Status Mapping	6	code
react_frontend	Frontend	6	code
rest_endpoints	API Design	6	document
shared_components	Frontend	5	code

The **`cargo_service_layer`** is the single most critical node — every page depends on it. The **`status_mapper`** is the most technically complex.

Edge Analysis

Strongest Relations (EXTRACTED, confidence = 1.0)

43 edges are directly extracted from the codebase (89.6%). These include: - `depends_on` — frontend layer hierarchy - `contains` — mapper → maps, arch → adapter - `implements` — adapter interface → carrier implementations - `uses` — pages → service layer, forms → pages - `part_of` — components → frontend framework - `maps_to` — status mapper → unified status model

Inferred Relations (confidence < 1.0)

5 edges are semantically inferred (10.4%): - `complements` — status_mapper ↔ carrier_adapter (0.85) - `crosscuts` — i18n_system → react_frontend (0.95) - `links_to` — dashboard → shipments (0.95), shipments → tracking (1.0) - `competes_with` — DHL ↔ FedEx (0.85), UPS ↔ Chronopost (0.75) - `required_by` — auth_system → endpoints (0.85)

Orphan & Overlap Analysis

Orphans: 0 — all nodes have at least 1 edge.
Overlaps: The `auth_system` node bridges Community 2 (Frontend) and Community 3 (API Design), as authentication crosscuts both frontend routing and backend endpoint security.

Key Architectural Insights

1. **Pluggable Carrier Architecture** — Adding a 5th carrier requires: (a) implement adapter interface, (b) add status map entries, (c) add mock data rows. No changes to existing code.

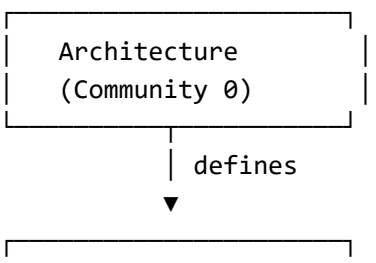
2. **Status Mapping is the Core Differentiator** — 3-tier matching with regex means the system handles real-world carrier status variations (typos, locale differences, formatting). Turkish carrier status names are supported via regex patterns.

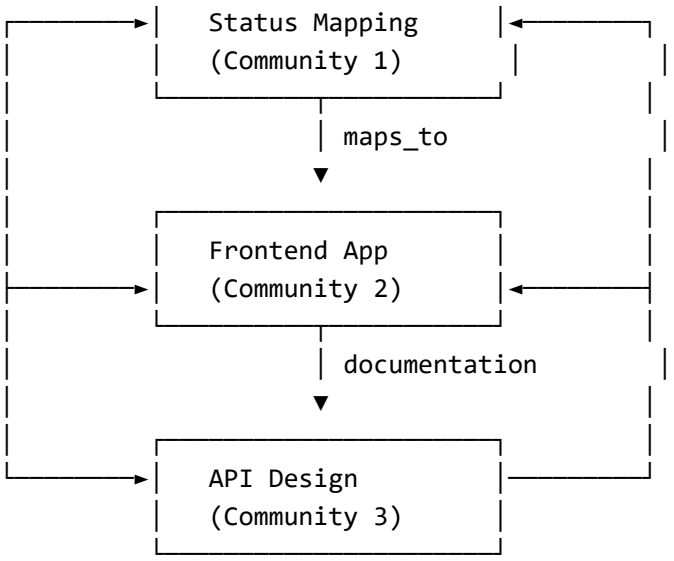
3. **Service Layer as God Node** — The single `cargoService.ts` module is the bottleneck. Consider splitting into domain-specific services (shipmentService.ts, carrierService.ts, etc.) for better maintainability.

4. **Frontend Outweighs Backend** — 21 of 40 nodes (52.5%) are frontend-related. The “backend” is currently mock data, but the architecture is ready for real API integration by swapping `cargoService.ts` implementations.

5. **Documentation is Centralized** — 3 service documentation files (EN, FR, color-coded) cover the same API spec, creating potential sync drift. Consider auto-generating from a single source.

Community Cross-Reference





File Details

Core Documents

File	Purpose
CARGO-SERVICE-ARCHITECTURE.md	Full backend architecture plan
service_cargo.md	API endpoint docs (French)
cargo_service.md	API endpoint docs (English)
cargo_service_api.md	Color-coded API reference

Frontend Source

File/Dir	Purpose
frontend/src/types/index.ts	14 TypeScript interfaces
frontend/src/data/mockData.ts	6 entity mock stores
frontend/src/services/statusMapper.ts	Status mapping engine (4 carriers)
frontend/src/services/cargoService.ts	20+ mock API functions
frontend/src/hooks/useCargoService.ts	useApi, useApiById, useMutation, useLoading
frontend/src/context/AuthContext.tsx	JWT mock auth state
frontend/src/i18n/	EN/FR/TR translation system
frontend/src/pages/	9 page components
frontend/src/components/	10+ shared components
frontend/src/components/forms/	4 form components

Report generated by Graphify knowledge graph pipeline.